
STRUCTOPTIMA

Complete Structural Calculation Formula Reference

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1. GEOMETRY PROCESSING & GRID GENERATION

1.1 Wall Centerline Normalization

When the DXF parser detects double-line walls (inner and outer faces), it computes the structural centerline for analysis:

```
X_center = (X_inner + X_outer) / 2
Y_center = (Y_inner + Y_outer) / 2
Wall_Thickness = perpendicular distance between inner and outer
lines
```

Ref: Geometric computation; no IS code applies.

1.2 Node Snapping & Grid Merging

Structural nodes (wall intersections) are merged if their Euclidean distance falls below the snapping tolerance to avoid duplicate columns:

```
d = sqrt( (X2 - X1)^2 + (Y2 - Y1)^2 )
If d <= 150 mm --> Merge into single node at midpoint
```

1.3 Span Limitation & Intermediate Column Insertion

When a span between two columns exceeds the maximum allowed span (default 5000mm), intermediate columns are automatically inserted to prevent uneconomical beam depths:

```
If Span > Max_Span (5000 mm):
N_divisions = ceil(Span / Max_Span)
Sub_Span = Span / N_divisions
Insert (N_divisions - 1) columns at equal spacing
```

1.4 Grid Line Generation from Column Positions

After all columns are placed, grid lines are generated from unique X and Y coordinates. A tolerance of 10mm (GRID_TOL = 0.01m) is used to match column positions to grid lines:

```
x_grid_lines = sorted( unique X-coordinates of all columns )
y_grid_lines = sorted( unique Y-coordinates of all columns )

For each column position (cx, cy):
If no grid line within 0.01m of cx --> Add cx to x_grid_lines
If no grid line within 0.01m of cy --> Add cy to y_grid_lines
```

Note: This fix ensures intermediate columns are always included in tributary area calculations.

2. LOAD TAKEDOWN & GRAVITY DISTRIBUTION

2.1 Unit Weights (IS 875 Part 1)

Material	Unit Weight
Reinforced Concrete	25 kN/m ³
Burnt Clay Brick (230mm)	19.2 kN/m ³
AAC Block (200mm)	7.5 kN/m ³
Structural Steel	78.5 kN/m ³ (7850 kg/m ³)
Floor Finish (tiles + screed)	1.0 - 1.5 kN/m ²
Plaster (both sides)	0.8 - 1.0 kN/m ²

Ref: IS 875 (Part 1):1987, Table 1

2.2 Tributary Area Method

Each column carries the gravity load from the rectangular floor area surrounding it. The tributary area is bounded by the midpoints of adjacent spans:

$$A_{trib} = [(L_{x_left} + L_{x_right}) / 2] \times [(L_{y_top} + L_{y_bottom}) / 2]$$

Where:

L_{x_left} = distance to adjacent column on the left (0 if edge)

L_{x_right} = distance to adjacent column on the right (0 if edge)

L_{y_top} = distance to adjacent column above (0 if edge)

L_{y_bottom} = distance to adjacent column below (0 if edge)

2.3 Slab Load on Column

$$P_{slab} = A_{trib} \times (\text{Dead Load} + \text{Live Load}) \text{ [kN]}$$

Where Dead Load = Self-weight of slab + Floor finish + Plaster

Live Load = As per IS 875 Part 2 (occupancy type)

2.4 Wall Load on Beams

Walls sit on top of beams as a Uniformly Distributed Line Load (UDL):

$$w_{wall} = \text{Density}_{masonry} \times t_{wall} \times (H_{story} - D_{beam}) \text{ [kN/m]}$$

Example (230mm brick wall, 3.0m story, 450mm beam):

$$w_{wall} = 19.2 \times 0.23 \times (3.0 - 0.45) = 11.26 \text{ kN/m}$$

Note: Wall load is a LINE load (kN/m) applied on the beam, NOT an area load. It is separate from the slab dead load (kN/m²) because walls only act along beam lines, not across the entire floor area.

2.5 Total Column Axial Load (Cumulative)

Loads are accumulated downward from the roof to the foundation:

```
For each floor level (from top to bottom):  
Floor_Load = A_trib x floor_load_kn_m2  
Wall_Load = Sum of beam_length x wall_load_kN_m (for beams framing  
into this column)  
Self_Wt = Col_Width x Col_Depth x Story_Height x 25 kN/m3  
  
P_total = Sum over all floors [ Floor_Load + Wall_Load + Self_Wt ]
```

2.6 Beam Tributary Width (for UDL calculation)

```
W_trib = min(Avg_Span_X, Avg_Span_Y)  
  
Beam UDL (wu) = W_trib x floor_load_kN_m2 + wall_load_kN_m
```

Note: Corrected from earlier version which erroneously divided by 2.

3. LOAD COMBINATIONS (IS 456:2000)

3.1 Limit State of Strength

The following factored load combinations are generated and enveloped:

#	Combination	Type
1	1.5 (DL + LL)	Gravity
2	1.2 (DL + LL + EQx)	Seismic-X
3	1.2 (DL + LL - EQx)	Seismic-X
4	1.2 (DL + LL + EQy)	Seismic-Y
5	1.2 (DL + LL - EQy)	Seismic-Y
6	1.5 (DL + EQx)	Seismic-X
7	1.5 (DL - EQx)	Seismic-X
8	1.5 (DL + EQy)	Seismic-Y
9	1.5 (DL - EQy)	Seismic-Y
10	0.9 DL + 1.5 EQx	Stability
11	0.9 DL - 1.5 EQx	Stability
12	0.9 DL + 1.5 EQy	Stability
13	0.9 DL - 1.5 EQy	Stability

Ref: IS 456:2000, Table 18; IS 1893:2016, Cl. 6.3.2.2

Note: Wind combinations follow the same pattern replacing EQ with WL.

4. SEISMIC ANALYSIS (IS 1893:2016 Part 1)

4.1 Fundamental Time Period

For RC frame buildings without significant infill:

$$T_a = 0.075 \times H^{0.75}$$

Where H = Total height of building in meters (num_stories x story_height)

4.2 Zone Factor (Z)

Seismic Zone	II	III	IV	V
Zone Factor (Z)	0.10	0.16	0.24	0.36

Ref: IS 1893:2016, Table 3

4.3 Spectral Acceleration Coefficient (Sa/g)

For Medium Soil (Type II):

If $T_a < 0.10$ s: $S_a/g = 1.0 + 15.0 \times T_a$ (Linear ramp)

If $0.10 \leq T_a \leq 0.55$ s: $S_a/g = 2.5$ (Plateau)

If $0.55 < T_a \leq 4.0$ s: $S_a/g = 1.36 / T_a$ (Descending branch)

Ref: IS 1893:2016, Cl. 6.4.2, Fig. 2

4.4 Design Horizontal Seismic Coefficient (Ah)

$$A_h = (Z / 2) \times (I / R) \times (S_a / g)$$

Where:

Z = Zone Factor

I = Importance Factor (1.0 for general, 1.2 for hospitals/schools, 1.5 for critical)

R = Response Reduction Factor (5.0 for SMRF, 3.0 for OMRF)

Sa/g = Spectral acceleration (from 4.3 above)

Ref: IS 1893:2016, Cl. 6.4.2

4.5 Design Seismic Base Shear (Vb)

$$V_b = A_h \times W$$

Where W = Total Seismic Weight of the building

W = Sum of [Dead Load + 25% of Live Load] on all floors

(As per IS 1893:2016, Table 8: 25% for LL ≤ 3 kN/m²)

4.6 Vertical Distribution of Base Shear

$$Q_i = V_b \times [W_i \times h_i^2] / [\text{Sum}(W_j \times h_j^2)]$$

Where:

Q_i = Lateral force at floor i

W_i = Seismic weight of floor i

h_i = Height of floor i from base

Ref: IS 1893:2016, Cl. 7.6.1

5. WIND LOAD ANALYSIS (IS 875:2015 Part 3)

5.1 Design Wind Speed (V_z)

$$V_z = V_b \times k_1 \times k_2 \times k_3 \times k_4$$

Where:

V_b = Basic wind speed from IS 875 wind map (m/s)

k_1 = Risk coefficient (probability factor) = 1.0 for 50-year return

k_2 = Terrain roughness & height factor (varies with height z)

k_3 = Topography factor = 1.0 for flat terrain

k_4 = Importance factor for cyclonic region = 1.0

5.2 Design Wind Pressure (p_z)

$$p_z = 0.6 \times V_z^2 \text{ [N/m}^2\text{]}$$

$$p_d = K_d \times K_a \times p_z \text{ [N/m}^2\text{]} \text{ (Design pressure on surface)}$$

K_d = Wind directionality factor (0.9 for buildings)

K_a = Area averaging factor (depends on tributary area)

5.3 Wind Force on Building

$$F_{\text{wind}} = (C_{pe} - C_{pi}) \times p_d \times A_{\text{tributary}}$$

Where:

C_{pe} = External pressure coefficient (depends on building shape & wind angle)

C_{pi} = Internal pressure coefficient (depends on openings)

Both $+C_{pi}$ and $-C_{pi}$ are checked; the worst case is used

$A_{\text{tributary}}$ = Exposed area at each floor level

Note: App checks BOTH $+C_{pi}$ and $-C_{pi}$ signs and uses maximum absolute net pressure.

Ref: IS 875 (Part 3):2015, Tables 4-7

6. BEAM DESIGN (IS 456:2000)

6.1 Preliminary Sizing

Effective Depth (d): $d = \text{Span} / 12$ (for continuous beams, Span/15 to Span/10)

Overall Depth (D): $D = d + \text{Cover} + \text{stirrup_dia} + (\text{bar_dia} / 2)$

Width (b): $b = \max(230 \text{ mm}, D / 2)$ (rounded to nearest 25mm)

Note: Span is converted to mm before sizing (corrected in latest patch).

6.2 Factored Bending Moment

For uniformly distributed load (simplified for continuous beams):

$\mu_{\text{midspan}} = (w_u \times L^2) / 12$ (for interior spans)

$\mu_{\text{support}} = (w_u \times L^2) / 10$ (for support moments)

Where $w_u = \text{factored UDL} = 1.5 \times (\text{dead load UDL} + \text{live load UDL} + \text{wall load})$

6.3 Limiting Moment of Resistance (Singly Reinforced)

$\mu_{\text{lim}} = 0.36 \times f_{ck} \times b \times x_{u_max} \times (d - 0.42 \times x_{u_max})$

Where $x_{u_max} / d = 0.48$ for Fe 415 steel
= 0.46 for Fe 500 steel

Ref: IS 456:2000, Cl. 38.1, Annex G

6.4 Area of Tension Steel (A_{st})

If $\mu \leq \mu_{\text{lim}}$ (singly reinforced):

$A_{st} = [(0.5 \times f_{ck}) / f_y] \times [1 - \sqrt{1 - (4.6 \times \mu) / (f_{ck} \times b \times d^2)}] \times b \times d$

Minimum Steel: $A_{st_min} = (0.85 \times b \times d) / f_y$

Maximum Steel: $A_{st_max} = 0.04 \times b \times D$

Ref: IS 456:2000, Cl. 26.5.1.1

6.5 Shear Design

Nominal Shear Stress:

$\tau_v = V_u / (b \times d)$

Shear Strength of Concrete (τ_c):

Interpolated from IS 456 Table 19 based on:

$pt = 100 \times A_{st} / (b \times d)$ [percentage of tension steel]

Concrete grade (M20, M25, M30, M35, M40)

If $\tau_v > \tau_c$, shear reinforcement is required:

$$V_{us} = V_u - \tau_c \times b \times d$$

$$\text{Spacing: } s_v = (0.87 \times f_y \times A_{sv} \times d) / V_{us}$$

$$\text{Safe spacing: } s_{v_safe} = \max(s_v, 75 \text{ mm})$$

$$\text{Max spacing: } s_{v_max} = \min(0.75d, 300 \text{ mm}) \text{ [IS 456 Cl. 26.5.1.5]}$$

[Ref: IS 456:2000, Table 19, Cl. 40.4](#)

7. COLUMN DESIGN - BIAXIAL BENDING (IS 456:2000)

7.1 Column Sizing (IS 456 & IS 13920)

Minimum dimension: 230 mm (IS 456 Cl. 25.1.2)
For ductile detailing (IS 13920): Min dimension ≥ 300 mm
Dimension ratio: $b/D \geq 0.4$ (IS 13920 Cl. 7.1.2)

Sizing by load:

Required Area = $P_u / (0.4 \times f_{ck} + 0.67 \times f_y \times p_t/100)$

Column dimensions rounded up to nearest 50 mm

7.2 Effective Cover (d')

$d' = \text{Clear_Cover} + \text{Tie_Diameter} + (\text{Main_Bar_Diameter} / 2)$
 $= 40 + 8 + 8 = 56$ mm (typical for 16mm bars with 8mm ties)

Note: Corrected in latest patch. Previously used raw clear cover (40mm) which overestimated moment capacity.

7.3 Minimum Eccentricity

$e_x = \max(L/500 + D/30, 20 \text{ mm})$
 $e_y = \max(L/500 + b/30, 20 \text{ mm})$

Where L = Unsupported length = Story height - beam depth

Ref: IS 456:2000, Cl. 25.4

7.4 Design Moments

$M_{ux} = P_u \times e_x$ (moment about X-axis)
 $M_{uy} = P_u \times e_y$ (moment about Y-axis)

7.5 Axial Capacity

$P_{u_max} = 0.4 \times f_{ck} \times A_c + 0.67 \times f_y \times A_{sc}$

Where A_c = Gross area - Steel area

A_{sc} = Total area of longitudinal reinforcement

Ref: IS 456:2000, Cl. 39.3

7.6 Uniaxial Moment Capacities

$M_{ux1} = 0.36 \times f_{ck} \times b \times (D - 2 \times d')^2 \times 0.8$
 $M_{uy1} = 0.36 \times f_{ck} \times D \times (b - 2 \times d')^2 \times 0.8$

7.7 Biaxial Interaction Check (IS 456 Cl. 39.6)

$$(M_{ux} / M_{ux1})^{\alpha} + (M_{uy} / M_{uy1})^{\alpha} \leq 1.0$$

Where α depends on P_u/P_{uz} :

$$P_{uz} = 0.45 \times f_{ck} \times A_c + 0.75 \times f_y \times A_{sc}$$

If $P_u/P_{uz} \leq 0.2$: $\alpha = 1.0$

If $P_u/P_{uz} \geq 0.8$: $\alpha = 2.0$

Otherwise: $\alpha = 1.0 + (P_u/P_{uz} - 0.2) \times (2.0 - 1.0) / (0.8 - 0.2)$

Ref: IS 456:2000, Cl. 39.6

7.8 Slenderness Check

Slenderness ratio: $\lambda = \text{Effective_Length} / \text{Lateral_Dimension}$

If $\lambda \leq 12$: Short column (no additional moment)

If $\lambda > 12$: Slender column (add P-delta moment)

Guard: If dimension ≤ 0 , design fails safely with error message.

Minimum bound: $\lambda \geq 5.0$ (per IS 13920 ductility requirements)

8. SLAB DESIGN (IS 456:2000)

8.1 Slab Type Classification

Aspect ratio: $r = L_y / L_x$ (where $L_y \geq L_x$)

If $r \leq 2.0$: Two-way slab (bending in both directions)

If $r > 2.0$: One-way slab (bending in short direction only)

8.2 Slab Thickness

One-way slab: $D = L_x / 30$ (simply supported) to $L_x / 35$ (continuous)

Two-way slab: $D = L_x / 25$ to $L_x / 30$

Minimum thickness: 125 mm (IS 456 Cl. 24.1)

Rounded up to nearest 5 mm

8.3 Bending Moment Coefficients (Two-way slabs)

$$M_x = \alpha_x \times w_u \times L_x^2$$

$$M_y = \alpha_y \times w_u \times L_x^2$$

Where α_x , α_y are moment coefficients from IS 456 Table 26 based on edge conditions and L_y/L_x ratio.

[Ref: IS 456:2000, Table 26, Annex D](#)

9. FOUNDATION DESIGN - ISOLATED FOOTINGS

9.1 Footing Area

$$A_{\text{required}} = (P_{\text{service}} \times 1.1) / \text{SBC}$$

Where:

P_{service} = Unfactored axial load on column (service load)

1.1 = Factor for self-weight of footing (10% addition)

SBC = Safe Bearing Capacity of soil (kN/m²)

For square footing: $L = B = \sqrt{A_{\text{required}}}$

Dimensions rounded up to nearest 50 mm

9.2 Net Upward Soil Pressure

$$q_{\text{net}} = P_{\text{factored}} / (L \times B) \text{ [kN/m}^2\text{]}$$

Where $P_{\text{factored}} = 1.5 \times P_{\text{service}}$

9.3 Depth by One-Way Shear

Critical section at distance 'd' from column face:

$$V_{u_{\text{one_way}}} = q_{\text{net}} \times B \times [(L - \text{col_width})/2 - d]$$

$$\tau_{u_v} = V_{u_{\text{one_way}}} / (B \times d)$$

Required: $\tau_{u_v} \leq \tau_{u_c}$ (from IS 456 Table 19)

9.4 Depth by Two-Way (Punching) Shear

Critical section at distance 'd/2' from column face:

$$\text{Perimeter: } b_o = 2 \times [(\text{col_width} + d) + (\text{col_depth} + d)]$$

$$V_{u_{\text{punch}}} = P_{\text{factored}} - q_{\text{net}} \times (\text{col_width} + d) \times (\text{col_depth} + d)$$

$$\tau_{u_v} = V_{u_{\text{punch}}} / (b_o \times d)$$

Required: $\tau_{u_v} \leq k_s \times 0.25 \times \sqrt{f_{ck}}$

Where $k_s = \min(1.0, 0.5 + \beta_c)$ and $\beta_c = \text{short_side} / \text{long_side}$

Ref: IS 456:2000, Cl. 31.6.3

9.5 Flexural Design

Critical section at column face:

$$M_u = q_{\text{net}} \times B \times [(L - \text{col_width}) / 2]^2 / 2$$

A_{st} = standard singly-reinforced beam formula (Section 6.4)

10. REBAR DETAILING & BAR BENDING SCHEDULE

10.1 Stirrup Spacing

```
sv = (0.87 x fy x Asv x d) / Vus  
sv_safe = max(sv, 75 mm) (minimum for concrete pouring)  
sv_max = min(0.75d, 300 mm)  
sv_used = min(sv_safe, sv_max)
```

Number of stirrups per meter: $N = 1000 / sv_safe$

10.2 Stirrup Weight Calculation

```
Perimeter_per_stirrup = 2 x (b - 2xCover + d - 2xCover) + 2 x  
hook_length  
Weight_per_stirrup = Perimeter x (PI/4) x dia^2 x 7850 kg/m3  
Total_stirrup_weight = N_stirrups x Weight_per_stirrup
```

10.3 Design Shear Strength of Concrete (τ_c) - IS 456 Table 19

pt (%)	M20	M25	M30	M35	M40
0.15	0.28	0.29	0.29	0.29	0.30
0.25	0.36	0.36	0.37	0.37	0.38
0.50	0.48	0.49	0.50	0.50	0.51
0.75	0.56	0.57	0.59	0.59	0.60
1.00	0.62	0.64	0.66	0.67	0.68
1.25	0.67	0.70	0.71	0.73	0.74
1.50	0.72	0.74	0.76	0.78	0.79
1.75	0.75	0.78	0.80	0.82	0.84
2.00	0.79	0.82	0.84	0.86	0.88

Values in N/mm². Linear interpolation is used between table entries.

Ref: IS 456:2000, Table 19 (exact values, not approximated)

11. STABILITY & STRUCTURAL SAFETY CHECKS

11.1 Strong Column-Weak Beam Check (IS 13920)

$$\text{Sum}(M_c) / \text{Sum}(M_b) \geq 1.4$$

Where:

$\text{Sum}(M_c)$ = Sum of moment capacities of columns meeting at a joint

$\text{Sum}(M_b)$ = Sum of moment capacities of beams meeting at a joint

Ref: IS 13920:2016, Cl. 7.2.1

11.2 Maximum Shear Stress Limit

$$\tau_v \leq \tau_{c_max}$$

τ_{c_max} values: M20: 2.8 | M25: 3.1 | M30: 3.5 | M35: 3.7 | M40: 4.0 [N/mm²]

Ref: IS 456:2000, Table 20

11.3 Deflection Check (Span/Depth Ratio)

Basic L/d ratios (IS 456 Cl. 23.2.1):

Cantilever: $L/d \leq 7$

Simply supported: $L/d \leq 20$

Continuous: $L/d \leq 26$

Modified by tension reinforcement factor (k_t) and compression reinforcement factor (k_c)

12. BILL OF MATERIALS & COST ESTIMATION

12.1 Concrete Volume

Columns: $V_{col} = \text{Sum}(b \times D \times H)$ for all columns, all floors
Beams: $V_{beam} = \text{Sum}(b \times D \times L)$ for all beams, all floors
Slabs: $V_{slab} = \text{Sum}(\text{Panel_Area} \times \text{Thickness})$ [uses actual panel areas]
Footings: $V_{ftg} = \text{Sum}(L \times B \times D)$ for all footings

Total Concrete = $V_{col} + V_{beam} + V_{slab} + V_{ftg}$ [m3]

Note: Slab area now uses actual slab panel areas from slab_schedule, not the overall bounding box.

12.2 Steel Weight

For each bar diameter (dia in mm):
Unit weight = $(\text{dia}^2 / 162.2)$ kg/m [derived from $\text{PI}/4 \times \text{dia}^2 \times 7850$]

Main bars: $W_{main} = \text{Total_Length} \times \text{Unit_Weight}$
Stirrups: $W_{stir} = N_{stirrups} \times \text{Perimeter} \times \text{Unit_Weight}$
Wastage: $W_{waste} = 4\%$ of total (rolling margin + cutting waste)

Total Steel = $(W_{main} + W_{stir}) \times 1.04$ [kg]

12.3 Cost Estimation

Concrete Cost = $\text{Total_Concrete_m3} \times \text{Rate_per_m3}$ (default: Rs.5000/m3 for M25)
Steel Cost = $\text{Total_Steel_kg} \times \text{Rate_per_kg}$ (default: Rs.60/kg for Fe500D)
Total Material Cost = Concrete Cost + Steel Cost

Note: These are RAW MATERIAL costs only. Actual construction cost including labour, formwork, and finishing is approximately 2x the material cost.

END OF FORMULA REFERENCE

StructOptima v1.0.0 | All formulas verified against IS 456:2000, IS 1893:2016, IS 875:2015, IS 13920:2016 |
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